From Proxy Transport to Open Models

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Phase 6 actual-model validation for the Metric Stack of Concern
Research-Derived Experiments - budgeted Modal L4 sweep

Abstract

Phase 5 used proxy harnesses to decide whether expensive validation was warranted. Phase 6 runs that heavier tier directly: public decoder language models are scored by hidden-state geometry and action logprob margins, while public frozen sentence encoders are tested with a post-hoc value-weighted metric deformation. The result is deliberately bounded. It is actual open-model evidence, not human behavioral evidence or foundation-model finetuning evidence.

1. Question

The question is whether the Phase 5 proxy transport signals survive contact with actual open LMs and frozen encoders. For language, the validation asks whether a hidden concern axis predicts held-out help-versus-wait logprob margins and whether concern cues move those margins more than matched controls. For encoders, it asks whether a value direction trained from template-A text improves held-out value-neighbor retrieval without random-label lift or off-target collapse.

preset	full
models	5 public HF models
gpu	L4
rows	5
claim level	actual open-LM and frozen-encoder validation result

Table 1. Suite manifest.

2. Real-Model Gates

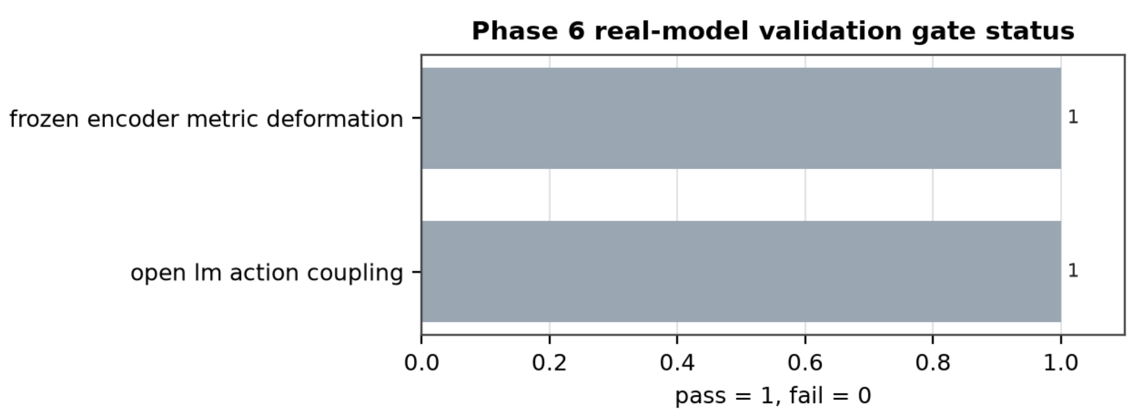

Figure 1. Pass/fail status for Phase 6 real-model validation gates.

Track	Status	Primary result	Allowed claim
Open LMs	PASS	models=3; r=0.340; lift=1.064; AUC=0.688	LM text signal
Frozen encoders	PASS	margin lift=0.409; random=0.008; AUC=1.000	metric margin

Table 2. Phase 6 validation gates and allowed claims.

3. Open Language Models

Each decoder LM is treated as a frozen measurement instrument. The suite builds a concern axis from train prompts, evaluates held-out prompts, and scores the average logprob margin between help and wait continuations. This tests text-model coupling only: a positive result means the open LM's hidden states and next-token distribution carry the measured concern signal, not that the model would act in the world.

LM	r	gap	lift	AUC	cue
DistilGPT2	0.454	0.016	0.703	0.750	-0.124
Pythia-70M	-0.035	0.012	0.000	0.500	-3.250
Qwen2.5-0.5B	0.602	0.153	2.488	0.812	1.312

Table 3. Public decoder-LM hidden geometry and logprob results.

4. Frozen Encoders

Each sentence encoder is frozen. The suite estimates a value direction from training phrases, then deforms held-out retrieval with a positive value-kernel term. The control is the same deformation after randomizing value scores. This isolates whether the metric story survives real encoder embeddings, not whether the encoder learned concern.

Encoder	raw P	def P	m-lift	rand	AUC	drift
MiniLM-L6	1.000	1.000	0.377	0.006	1.000	0.000
BGE-small	1.000	1.000	0.441	0.011	1.000	0.000

Table 4. Public frozen-encoder metric deformation results.

5. Interpretation

The strictest interpretation is operational. Passing gates promote the relevant mechanism from proxy-compatible to actual-model-compatible. Failing or weak gates demote that path back to hypothesis status while preserving the exact model rows as future baselines.

Gate snapshot: language pass=PASS; encoder pass=PASS; overall=PASS.

6. Discovery-Regime Audit

Old regime: Phase 5 proxy transport. Transition: public model weights replace proxy weights while preserving predeclared controls, budget caps, and archiveable artifacts. Promoted evidence is restricted to actual open-model logprob/hidden-state measurements and frozen-encoder metric deformation. Rejected alternatives remain explicit: random deformation, weak cue specificity, off-target drift, and model-loading failures.

References

Brown, J. From Controlled Concern Geometry to Foundation Models. Research-Derived Experiments, 2026.

Brown, J. Learning the Missing Conditions. Research-Derived Experiments, 2026.

Hugging Face model cards for distilgpt2, EleutherAI/pythia-70m-deduped, Qwen/Qwen2.5-0.5B-Instruct, all-MiniLM-L6-v2, and BAAI/bge-small-en-v1.5.